Section 5: Chemistry in society

a) Extraction and uses of metals

b) Crude oil

c) Synthetic polymers

d) The industrial manufacture of chemicals

a) Extraction and uses of metals

Students will be assessed on their ability to:

5.1 explain how the methods of extraction of the metals in this section are related to their positions in the reactivity series

5.2 describe and explain the extraction of aluminium from purified aluminium oxide by electrolysis, including:

- i the use of molten cryolite as a solvent and to decrease the required operating temperature
- ii the need to replace the positive electrodes
- iii the cost of the electricity as a major factor

5.3 write ionic half-equations for the reactions at the electrodes in aluminium extraction

5.4 describe and explain the main reactions involved in the extraction of iron from iron ore (haematite), using coke, limestone and air in a blast furnace

5.5 explain the uses of aluminium and iron, in terms of their properties.

b) Crude oil

Students will be assessed on their ability to:

5.6 recall that crude oil is a mixture of hydrocarbons

5.7 describe how the industrial process of fractional distillation separates crude oil into fractions

5.8 recall the names and uses of the main fractions obtained from crude oil: refinery gases, gasoline, kerosene, diesel, fuel oil and bitumen

5.9 describe the trend in boiling point and viscosity of the main fractions

5.10 recall that incomplete combustion of fuels may produce carbon monoxide and explain that carbon monoxide is poisonous because it reduces the capacity of the blood to carry oxygen

5.11 recall that, in car engines, the temperature reached is high enough to allow nitrogen and oxygen from air to react, forming nitrogen oxides

5.12 recall that fractional distillation of crude oil produces more long-chain hydrocarbons than can be used directly and fewer short-chain hydrocarbons than required

5.13 describe how long-chain alkanes are converted to alkenes and shorter-chain alkanes by catalytic cracking, using silica or alumina as the catalyst and a temperature in the range of 600–700°C.

c) Synthetic polymers

Students will be assessed on their ability to:

5.14 recall that an addition polymer is formed by joining up many small molecules called monomers

5.15 draw the repeat unit of addition polymers, including poly(ethene), poly(propene) and poly(chloroethene)

5.16 deduce the structure of a monomer from the repeat unit of an addition polymer

5.17 recall that nylon is a condensation polymer

5.18 understand that the formation of a condensation polymer is accompanied by the release of a small molecule such as water or hydrogen chloride

5.19 recall the types of monomers used in the manufacture of nylon

5.20 draw the structure of nylon in block diagram format.

d) The industrial manufacture of chemicals

Students will be assessed on their ability to:

5.21 recall that nitrogen from air, and hydrogen from natural gas or the cracking of hydrocarbons, are used in the manufacture of ammonia

5.22 describe the manufacture of ammonia by the Haber process, including the essential conditions:

- i a temperature of about 450°C
- ii a pressure of about 200 atmospheres
- iii an iron catalyst

5.23 understand how the cooling of the reaction mixture liquefies the ammonia produced and allows the unused hydrogen and nitrogen to be recirculated

5.24 recall the use of ammonia in the manufacture of nitric acid and fertilisers

5.25 recall the raw materials used in the manufacture of sulfuric acid

5.26 describe the manufacture of sulfuric acid by the contact process, including the essential conditions:

- **i a temperature of about 450 °C**
- **ii a pressure of about 2 atmospheres**
- **iii a vanadium(V) oxide catalyst**

5.27 recall the use of sulfuric acid in the manufacture of detergents, fertilisers and paints

5.28 describe the manufacture of sodium hydroxide and chlorine by the electrolysis of concentrated sodium chloride solution (brine) in a diaphragm cell

5.29 write ionic half-equations for the reactions at the electrodes in the diaphragm cell

5.30 recall important uses of sodium hydroxide, including the manufacture of bleach, paper and soap; and of chlorine, including sterilising water supplies and in the manufacture of bleach and hydrochloric acid.